

TECHNICAL SETUP GUIDE

Day 12: Installing Anaconda, Jupyter Notebook & Google Colab Workflows

1. Executive Summary & Overview

What This Session Covers:

- Setting up a professional environment to write and run Machine Learning code.
- How to use the three most popular tools in Data Science: Anaconda (Local), Kaggle (Cloud), and Google Colab (Cloud).
- Why Virtual Environments are critical for protecting and managing your project libraries.
- A secret trick to download giant datasets directly from Kaggle into Google Colab in seconds.

2. Local Workspace Setup: Anaconda & Jupyter Notebook

To run Machine Learning code on your own laptop, installing libraries one by one can be highly frustrating. The industry standard tool to solve this issue is **Anaconda**.

- **What is Anaconda?** It is a free platform that bundles together Python and hundreds of pre-installed data science libraries (like NumPy and Pandas). This saves you from running installation tasks manually.
- **Jupyter Notebook vs. Traditional IDEs:** Instead of writing code in a standard plain text file, Data Scientists use Jupyter Notebooks. It breaks down your code workspace into individual block cells. You can run one specific block of code at a time instead of executing the entire file from scratch.

JUPYTER NOTEBOOK

[Cell 1: Markdown] # My Title Block

[Cell 2: Python] import pandas as pd
 df = pd.read_csv("dataset.csv")

[Cell 3: Python] df.head()
 ▶ Output: Interactive Data Table

Jupyter Notebook supports two main cell modes: **Code cells** (to write and run Python lines) and **Markdown cells** (to write clean text notes, add bold subheadings, include pictures, or even write styled HTML tags to document your project journey).

3. Managing Workspaces with Virtual Environments

When you launch a basic Jupyter session out of the box, it connects to your "Base" environment where all your packaged libraries live together. While convenient at first, this approach breaks down when deploying real applications.



If you deploy your project to a cloud server, the hosting platform will ask for your environment blueprint file. If you are using the Base environment, the server will try to download hundreds of unnecessary libraries, expanding your file bundle size and creating software errors. Creating a clean virtual container box ensures you only install the precise packages your project actually needs.

Useful Terminal Commands (Anaconda Prompt)

- Create a new environment container:

```
conda create --name MyNewEnvironment python=3.9
```

- Turn on your container environment:

```
conda activate MyNewEnvironment
```

- Install a specific library inside the container:

```
conda install jupyter notebook numpy
```

4. Cloud Workspaces: Kaggle & Google Colab

If you have an older laptop or do not want to install large software packages locally, you can perform your Data Science work completely in the cloud using your web browser.

- **Kaggle Notebooks:** A free web platform that hosts thousands of open datasets. You can start a clean notebook directly in your browser attached to those datasets, completely bypassing local download requirements. You can easily share your code files and build up a data portfolio.
- **Google Colab:** Google's online version of Jupyter Notebook. It saves files automatically into your personal Google Drive account. More importantly, it gives you free access to powerful hardware accelerators like **GPUs (Graphics Processing Units)**, which are necessary for high-speed Deep Learning pipelines.

5. Advanced Workflow Trick: Direct Cloud Data Transfer

When working on large Deep Learning problems, you often find top-tier image files hosted on Kaggle, but you want to run your models using Google Colab's free GPU hardware. Moving a 2GB dataset down to your desktop and then dragging it up to Google Colab wastes hours of time and internet bandwidth. You can connect them directly through the cloud command-line API instead.

[Kaggle Servers] ————— Direct API Transfer —————> [Google Colab]
(Holds 2GB Dataset) (Runs Free Cloud GPU)

Step-by-Step Implementation:

1. Go to your Kaggle Account profile settings and click "**Create New API Token**". This automatically downloads a security profile key called `kaggle.json` to your local machine.
2. Open your Google Colab file browser tab and upload the `kaggle.json` file straight into the root workspace folder.
3. Create a code cell block and execute the directory copy configuration instructions:

```
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json
```

1. Navigate back to the target Kaggle dataset web page, click the action dots, and select "**Copy API Command**". Paste that line into your Colab notebook with an exclamation mark at the front to run the file download:

```
!kaggle datasets download -d developer-name/dataset-slug
```

1. Unzip the transferred file package directly inside your cloud environment memory layer:

```
!unzip dataset-name.zip
```

6. Strategic Conclusion

Learning how to switch environments smoothly keeps your data science work quick and agile. Use local Jupyter Notebooks when prototyping small files locally, and transition to Google Colab when training complex models that require high-speed cloud GPU processors.